

Praveen Rathod

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Summary

AI Engineer with hands-on experience building RAG-based question answering systems, codebase chat applications, data validation pipelines, and rule-based financial analysis tools. Interned at Vyne AI, where I worked on a fully offline, privacy-first AI assistant using LLaMA and Mistral LLMs with on-device inference, vector-based memory, and deterministic reasoning to deliver reliable, cloud-independent intelligent systems.

Work Experience

Vyne AI

May 2025- Oct 2025

AI ENGINEER

Remote-USA

- Worked on a fully offline, privacy-first AI assistant using on-device inference with optimized LLaMA and Mistral LLMs.
- Built local LLM inference pipelines integrated with vector-based memory for persistent, context-aware responses.
- Implemented rule-based reasoning layers to ensure deterministic behavior and reliable outputs without any cloud dependency.
- Contributed to backend workflows supporting offline execution, agent orchestration, monitoring, and overall system reliability.

Notable Projects

Document Intelligence & Semantic Search

Oct 2025-Dec 2025

RAG-BASED QUESTION ANSWERING SYSTEM

IIT-Dhanbad

- Designed and implemented a document-based Q&A system using a Retrieval-Augmented Generation (RAG) pipeline for knowledge retrieval.
- Built robust document ingestion, text chunking, and embedding workflows to support efficient semantic search over large datasets.
- Integrated vector-based retrieval mechanisms to supply relevant contextual information for accurate and grounded answer generation.
- Exposed the system through a REST API to deliver structured, scalable, and context-aware query responses.

Code Intelligence & Developer Productivity

June 2025-Aug 2025

CHAT WITH CODEBASE (RAG)

IIT-Dhanbad

- Developed a system that enables users to query, explore, and understand large GitHub codebases using natural language.
- Implemented semantic code chunking at function and class levels to enable fine-grained and context-aware retrieval.
- Generated, stored, and retrieved code embeddings using a vector database to support precise technical explanations.
- Built an interactive chat interface that generates accurate responses grounded directly in the underlying source code.

LLM-Based Structured Reasoning System

Oct 2024-Dec 2024

AI PROJECT PLANNING & GUIDANCE CHATBOT

IIT-Dhanbad

- Designed and implemented an LLM-powered chatbot that transforms a project idea into structured technical guidance.
- Engineered prompt orchestration and context templates for structured roadmaps, tool recommendations, and workflows.
- Implemented structured output generation for flow-style explanations and mind-map project architectures.
- Exposed the system via a backend API to support interactive and stateless chatbot-based usage.

Education

Indian Institute of Technology (ISM) Dhanbad

Dhanbad, India

BACHELOR OF TECHNOLOGY IN MINING ENGINEERING

2022 – 2026

- Relevant Coursework: C++ Language, Data Structures & Algorithms, Object-Oriented Programming

Skills

- **Languages**: C++ | Python | JavaScript | TypeScript
- **Full-Stack (MERN)**: MongoDB | Express.js | React | Node.js | API Integration | SQL (MySQL) | Pinecone
- **Concepts**: Data Structures and Algorithms | Object Oriented Programming | Operating Systems | DBMS
- **AI & LLM Systems**: LLaMA | Mistral | GPT Models | RAG | Embeddings | Prompt Engineering | LangChain | Langgraph
- **Technologies**: GCP | AWS | Git | GitHub | VS Code | Antigravity

Achievements

- Solved **400+ LeetCode problems in C++** with strong problem-solving skills in DSA, dynamic programming, graphs, and algorithms.
- Achieved a **3-Star rating on CodeChef** and solved **300+ competitive programming problems** across coding platforms.
- Participated in the official **Spike AI Hackathon** on AI Agents conducted at **IIT (ISM) Dhanbad**.

Extracurricular Activities

- Member, Student Startup Club, IIT (ISM) Dhanbad — actively involved in startup ideation and early-stage product discussions.
- Founder & Creator, AI for Future — built an AI-focused Instagram and Facebook platform to **160K+ followers** with **200M+ organic views**.
- Head of the Public Relations Team at **Srijan'25** and Head of Event & Media Team at **Khanan'24**, IIT (ISM) Dhanbad.